StableAML: Machine Learning for Behavioral Wallet Detection in Stablecoin Anti–Money Laundering on Ethereum

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THE STABLECOIN LAUNDERING PROBLEM

Stablecoins (USDT/USDC) have emerged as the preferred medium for illicit value transfer, accounting for over 84% of verified crypto fraud volumes in 2025 due to their liquidity and rapid settlement. While decentralized protocols increasingly adopt zero-knowledge proofs to obfuscate transaction graphs, centralized stablecoins remain critical "transparent choke points" for compliance.

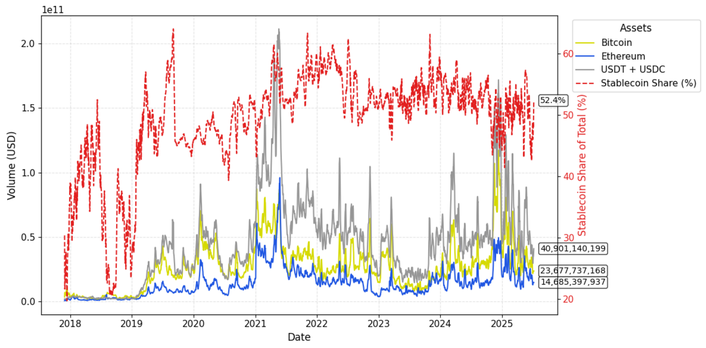

Figure 1: Historical transaction volumes for Bitcoin, Ethereum, and combined stablecoins.

The Challenge:

- Global illicit fund flows surpass $3.1 trillion annually. Blockchain mirrors financial crime stages but occurs much faster. In 2024, $51 billion was laundered through cryptocurrencies.
- While public ledgers provide transparency, privacy technologies (mixers) and cross-chain behaviors (swaps) fragment the transaction graph.
- Existing graph-based models struggle with this increasing sparsity, failing to link the placement and integration stages of money laundering.

Data Construction:
We constructed the **StableAML** dataset by filtering Ethereum logs specifically for USDT and USDC `Transfer` events.

- **Volume:** 334M transfers, 55M distinct addresses (2017-2025).
- **Classes:** Normal, Cybercrime (Exploits/Scams), and Blocklisted (Sanctioned/Frozen).

Field	Value
Transaction	`0xf8163c3d5ba77186ad4c6c93c4f3f92a88adb1b55cd0da34de2a44d33d4e20bd`
Sender	`0x654Fae4aa229d104CAbead47e56703f58b174bE4`
Recipient	`0x0000000000035B5e5ad9019092C665357240f594e`
Amount	1,092,761.61 USDT
Timestamp	2024-01-31 11:59:59

Table 1: Example of a real token transfer event extracted from the dataset.

Illicit Flow Visualization:

Figure 2: A real-world "layering" pattern: Funds move from Tornado Cash to a centralized exchange (Binance) via intermediary hops in less than 30 minutes.

METHODOLOGY: FEATURES VS. TOPOLOGY

We compare domain-informed feature engineering against graph representation learning.
1. The Failure of GNNs (GraphSAGE)

$$h_v^{(l)} = \sigma\left(W^{(l)} \cdot \left[h_v^{(l-1)} \| \frac{1}{|\mathcal{N}(v)|} \sum_{u \in \mathcal{N}(v)} h_u^{(l-1)}\right]\right)$$

where $\mathcal{N}(v)$ denotes the neighborhood. **Issue:** In stablecoin laundering, actors use swaps and bridges, effectively making $\mathcal{N}(v) \to \emptyset$ in a stablecoin-only graph. The sparsity (> 0.99) prevents effective embedding learning.

2. The Robustness of Tree Ensembles We utilize Gradient Boosting Machines (XGBoost, CatBoost) with a multi-class objective function to minimize aggregate cross-entropy loss:

$$\mathcal{L} = -\frac{1}{n}\sum_{k=1}^{K}\sum_{i=1}^{n}[y_{i,k}\log(p_{i,k}) + (1-y_{i,k})\log(1-p_{i,k})]$$

This allows the model to partition the feature space along discrete behavioral signals (e.g., 'receivedFromDex') rather than relying on topological continuity.

Feature Engineering (68 Dimensions): We extract 68 behavioral features categorized into:

- **Interaction:** Direct protocol usage (DeFi, CEX, Mixers).
- **Derived Network:** 2nd-degree exposure to high-risk entities (e.g., '2ndWithSwap').
- **Transfer-based:** Volume thresholds and temporal bursts.
- **Temporal and Direct Features:** Capture timing patterns and intrinsic properties.

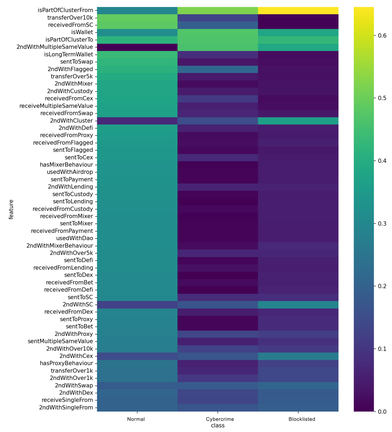

Figure 3: Feature Importance Heatmap: Distinct behavioral signatures. Cybercrime wallets rely on complex clustering features ('isPartOfCluster'), while Blocklisted show static, direct interaction patterns.

Distinct behavioral profiles :

- **Normal:** Defined by the absence of coordinated illicit signals. The model distinguishes benign activity purely by the lack of complex structures found in the suspicious categories.
- **Cybercrime:** Characterized by deep network features (e.g., `isPartOfCluster`) and indirect exposure. These structured signals reflect automated "layering," where syndicates utilize multi-hop routing and bridges to obscure origins.
- **Blocklisted:** Marked by direct interactions (`SentToSC`) and simpler paths. Unlike cybercrime, these patterns reflect constrained, post-enforcement behaviors and reduced mobility typical of sanctioned entities.

RESULTS AND INTERPRETABILITY

Performance Benchmarking: Tree ensembles significantly outperform Graph Neural Networks and Deep Learning baselines.

Model	AUROC	Precision/Acc.	F1 Score	Recall
CatBoost (Overall)	**0.9974**	**0.9857**	**0.9775**	**0.9757**
Class: Normal	0.9990	0.9927	0.9963	1.0000
Class: Cybercrime	0.9991	0.9873	0.9852	0.9831
Class: Blocklisted	0.9943	0.9579	0.9509	0.9509
XGBoost	0.9974	0.9851	0.9766	0.9760
DNN	0.9617	0.9192	0.8708	0.8504
GNN (GraphSAGE)	0.9683	0.8290	0.8048	0.7699

Table 2: Performance comparison. CatBoost results are expanded to show per-class metrics (Normal, Cybercrime, Blocklisted), highlighting robustness across all categories.

Behavioral Insights:

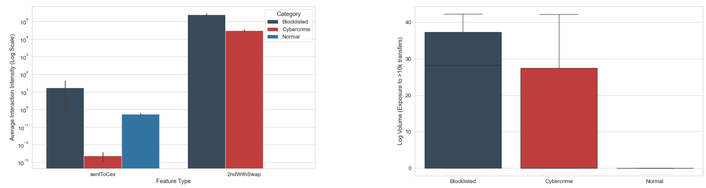

Figure 4: (Left) Interaction Profile: Illicit actors avoid CEXs ('sentToCex') and favor Swap protocols. (Right) High-Value Exposure: Cybercrime wallets have high 2nd-degree exposure to large transfers ('2ndWithOver10k'), unlike normal users.

TAKEAWAYS

- **Transparent Choke Points:** Centralized stablecoins remain the critical auditable layer in an increasingly private, zero-knowledge ledger economy.
- **Superior Robustness:** Tree ensembles significantly outperform Graph Neural Networks (Macro-F1 0.97 vs. 0.81) by overcoming transaction graph fragmentation.
- **Behavioral KYT:** The framework shifts compliance from identity (KYC) to action (KYT), using 68 engineered features to fingerprint laundering patterns without PII.
- **Typology Dissection:** Beyond binary detection, the model distinguishes the complex, high-velocity layering of cybercrime from the constrained footprints of sanctioned entities.
- **Regulatory Alignment:** Interpretable decision paths satisfy the strict auditability requirements of MiCA and the U.S. GENIUS Act while minimizing unjustified asset freezes.

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